Code No: 157AC **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2022
ADVANCED ALGORITHMS
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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JNTUH Used Papers

1.a) Show the outcome of different passes for sorting the following sequence of data using quick sort algorithm choose the first element as pivot. 8, 11, 3, 15, 6, 9, 12, 39.

b) What is dynamic programming? How can you solve the problem of matrix chain multiplication using dynamic programming explains in detail with example? [7+8]

2.a) Solve the following recurrence relation to give a tight upper bound using substitution method.
$T(n) = 4T(n/2)+n$

b) Determine Longest Common Subsequence (LCS) for input Sequences "ABCDGH" and "AEDFHR" using Dynamic programming. [7+8]

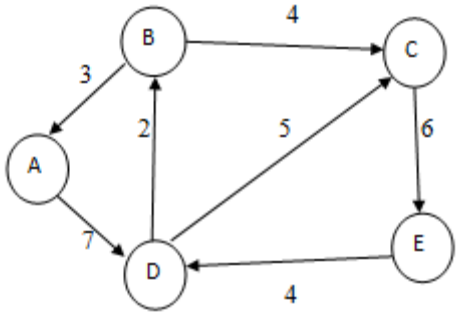
3.a) We have a set of n jobs to execute, each of which takes unit time. At a y time $T = 1, 2\ldots.$, we can execute exactly one jobs. Job i earns us a profit $g_i \rightarrow 0$ if and only if it is executed not later than the time d_1. Develop a greedy algorithm to solve the above problem. Run your algorithm for n=4 and the following values:-

i	1	2	3	4
g_i	50	10	15	30
d_i 2	2	1	2	7

b) Discuss about the algorithm and pseudo code to find minimum spanning tree using Prim's algorithm? [8+7]

4.a) What is an optimal Huffman code for the following set of frequencies, based on the first 8 Fibonacci numbers? a:1 b:1 c:2 d:3 e:5 f:8 g:13 h:21

b) Solve the following instance of the single source shortest path problem with vertex 'a' as the source. [7+8]



5.a) Explain in detail about Bitonic Sorting Network?

b) Explain in detail the implementation of Strassen's Matrix Multiplication with an example? [7+8]

6.a) Let n be an exact power of 2. Show how to construct an n-input, n-output comparison network of depth lg n in which the top output wire always carries the minimum input value and the bottom output wire always carries the maximum input value.

b) Illustrate LUP-Decomposition in detail? [8+7]

7.a) Show the comparisons the naive string matcher makes for the pattern $P = 0001$ in the text $T = 000010001010001$

b) Construct the string-matching automation for the pattern P = aabab and illustrate its operation on the text string T = aaababaabaababaab. [7+8]

8.a) Show that an algorithm that makes at most a constant number of calls to polynomial-time subroutines runs in polynomial time, but that a polynomial number of calls to polynomial-time subroutines may result in an exponential-time algorithm justify?

b) Professor Nixon proposes the following heuristic to solve the vertex-cover problem. Repeatedly select a vertex of highest degree, and remove all of its incident edges. Give an example to show that the professor's heuristic does not have a ratio bound of 2. [7+8]

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